ROBERTO GUAGNELLI GARCÍA

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Mechanical Engineering | Aeronautical Engineering Concentration | Design, Analysis & Manufacturing

Spanish / Mexican citizenship | EU work authorization | Graduating Jul 2027; available Aug 2027 | Open to relocation

EDUCATION

Tecnológico de Monterrey, Campus Estado de México Expected Jul 2027
B.S. Mechanical Engineering, Concentration in Aeronautical Engineering | **Academic average: 91/100**

INDUSTRY EXPERIENCE

GE Aerospace | Product Definition / Mechanical Design Intern Aug 2026 – Present
RISE program | Siemens NX, engineering drawings and mechanical interfaces

- Designed **70+ support brackets** and delivered **20+ harness-routing revisions**, considering assembly access, manufacturability, maintainability, clearances and cable bend radii.
- Translated harness architecture into Siemens NX 3D installations and product-definition drawings, resolving integration issues around surrounding systems and datum-based tolerances.
- Collaborate directly with vibration-analysis engineering teams in **Mexico and the United States**, incorporating vibration and fatigue considerations into design iterations.

GE Aerospace | Lean Summer Experience Intern Jun – Jul 2026
Integrated Performance & Operability (IP&O)

- Developed planning tools, KPI dashboards and Python automations with program leaders; meeting leaders measured approximately **20% shorter meetings** after process changes.

ENGINEERING LEADERSHIP & AIRCRAFT DEVELOPMENT

Borregos CEM AeroDesign | Tecnológico de Monterrey Sep 2023 – Present
Engineering & Leadership Advisor (2026–present); Design Captain / Engineering Coordinator (2024–2026); Manufacturing Coordinator (2023–2024)

- Led **five designers** and coordinated engineering interfaces within a **40+ person team**; helped deliver **three aircraft built and flown**. Modeled **300+ CAD parts per aircraft** within 400+ component assemblies.
- **Wingull (2025):** Designed removable joints for a 4.2 m wing with a 1.2 m transport constraint; supported an 18 kg distributed-load test and subsequent payload-carrying flight.
- **Latías (2026):** Modeled longitudinal flight dynamics in Python and XFLR5 (stability derivatives, state-space eigenvalues, phugoid and short-period modes); revised the loaded-CG target to recover modeled stability.
- **Hyperion (2024):** Coordinated eight builders and designed/fabricated spring-loaded aluminum nose landing gear. Team earned **2nd overall** at SAE Aero Design México; aircraft completed competition flights.
- **SAE AeroDesign 2027 (ongoing):** Co-developing the next aircraft with the Design Captain, supporting aerodynamic configuration, stability analysis and Python takeoff-performance modeling.

SELECTED TECHNICAL PROJECTS

Wing Aerodynamics | CFD and Design Studies 2026

- Built steady/transient wing simulations in **ANSYS Fluent** (CAD, meshing, SST k–ω, flow visualization) and compared aerodynamic trends with **XFLR5** to inform wing-incidence and tip-design decisions.

Quadrotor Aerodynamics & Arm Structural Analysis 2026

- Modeled hovering propeller flow and rotor interaction in **ANSYS Fluent**; studied a 7075-T6 aluminum motor arm in **ANSYS Mechanical** under thrust and bending loads, assessing deformation and angular deflection.

BIC Extrusion Cutter | Mechanism Prototype 2026

- Co-developed and assembled an instrumented single-motor cutting prototype with Geneva indexing, a gear reduction and synchronized double-blade cutting, using CAD, laser cutting, 3D printing and machining.

Extruder Screw Failure Analysis | Materials Investigation 2026

- Investigated a fractured tool-steel component using hardness tests, metallography, SEM/EDS and torsional stress analysis to assess competing failure mechanisms.

TECHNICAL SKILLS & CREDENTIALS

Design / analysis: Siemens NX, Teamcenter, SOLIDWORKS (**CSWP**), ANSYS Fluent / Mechanical, XFLR5, AVL; CAD, GD&T, CFD, FEA and flight dynamics.

Programming / fabrication: Python, MATLAB, Excel, Power BI; composites, 3D printing, laser cutting, machining, sheet-metal forming and aluminum welding.

Languages: Spanish (native), English (C2, IELTS 8.5), German (A2).

Additional digital production: 100+ 3D models/textures for Razzleberries' licensed Hello Kitty content, following brand guidelines; 25+ detailed Bloodborne-inspired models. Built animations and interactive behaviors with Blender, Blockbench and JSON.